Offline Wi-Fi Communication Between Raspberry Pi and Laptop Using ESP32

Project Objective:

The objective of this project is to establish offline wireless communication between a laptop and a Raspberry Pi using the Wi-Fi capability of an ESP32 module. The ESP32 acts as a local Wi-Fi access point, allowing both devices to connect to the same network without requiring internet access. Using Python socket programming, the laptop sends messages to the Raspberry Pi through a specific port over the local network. This project demonstrates how devices can communicate and exchange data using IP networking in an IoT environment.

Hardware Required:

Raspberry Pi

ESP32 Wi-Fi Module

Laptop

USB Cable

Software Used:

Arduino

Raspberry pi Imager

Project Architecture

In this system, the ESP32 is configured in Access Point mode, where it creates a local Wi-Fi network. Both the Raspberry Pi and the laptop connect to this network. Once connected, they become part of the same local network and can communicate without internet access.

The Raspberry Pi runs a server program using Python socket programming, while the laptop runs a client program. The laptop can send messages to the Raspberry Pi, and the Raspberry Pi can also send responses back to the laptop through the same connection. This enables two-way communication between both devices.

Procedure:

- **Configure the ESP32 as a Wi-Fi Access Point**
  Program the ESP32 to create its own Wi-Fi network with a specific SSID and password so that other devices can connect to it.

- **Connect Devices to the ESP32 Network**
  Connect the laptop and the Raspberry Pi to the Wi-Fi network created by the ESP32.

- **Find the Raspberry Pi IP Address**

- **Create the Server Program on Raspberry Pi**
  Write a socket server program using Python that listens for incoming connections on a specific port.

- **Run the Server Program**
  Execute the server program on the Raspberry Pi so that it waits for a connection request from the laptop.

- **Create the Client Program on the Laptop**
  Write a Python socket client program on the laptop that connects to the Raspberry Pi using its IP address and port number.

- **Send Data from Laptop to Raspberry Pi**
  Run the client program on the laptop to send messages through the ESP32 Wi-Fi network to the Raspberry Pi.

- **Receive and Display Data**
  The Raspberry Pi receives the message and displays it in the terminal. The Raspberry Pi can also send responses back to the laptop, enabling two-way communication.

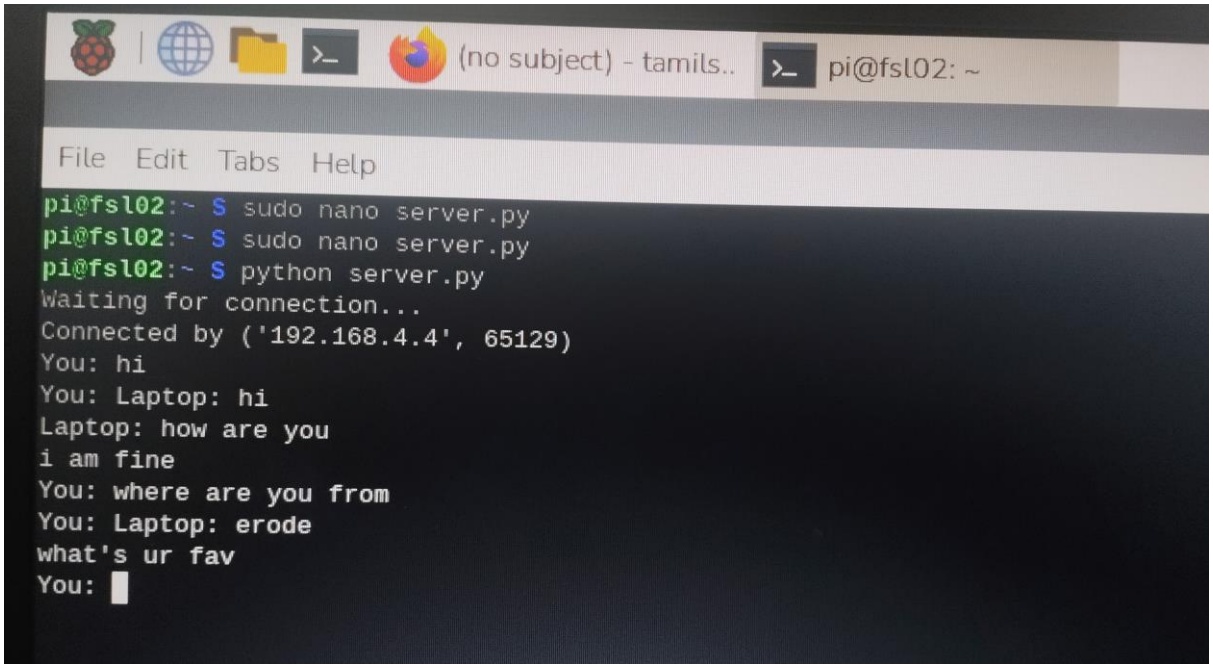
Raspberry pi Code:

```
import socket
import threading

host = '0.0.0.0'
port = 5000

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((host, port))
server.listen(1)

print("Waiting for connection...")
conn, addr = server.accept()
print("Connected by", addr)

# Function to receive messages
def receive():
```

```
    while True:
        data = conn.recv(1024).decode()
        if data.lower() == "exit":
            print("Laptop ended the chat")
            break
        print("\nLaptop:", data)

# Function to send messages
def send():
    while True:
        msg = input("You: ")
        conn.send(msg.encode())
        if msg.lower() == "exit":
            break

threading.Thread(target=receive).start()
threading.Thread(target=send).start()
```

Laptop Code:

```
import socket
import threading

host = "192.168.4.3"   # Raspberry Pi IP
port = 5000

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((host, port))

print("Connected to Raspberry Pi")

# Receive messages
def receive():
    while True:
        data = client.recv(1024).decode()
        if data.lower() == "exit":
            print("Raspberry Pi ended the chat")
            break
        print("\nRaspberry Pi:", data)

# Send messages
def send():
    while True:
        msg = input("You: ")
        client.send(msg.encode())
        if msg.lower() == "exit":
            break

threading.Thread(target=receive).start()
threading.Thread(target=send).start()
```

Output:

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Tamilselvan R\OneDrive\ras> python client.py
Connected to Raspberry Pi
You: Raspberry Pi: hi
hi
You: how are you
You: Raspberry Pi: i am fine
Raspberry Pi: where are you from
erode
You: Raspberry Pi: what's ur fav
```